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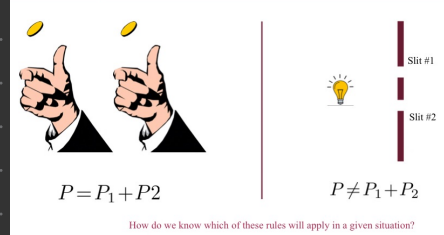
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• **COIN**: WE CAN CHECK WHICH SIDE IS FACING (HEAD OR TAIL)

• **SPLIT**: WE CANNOT KNOW WHICH SLIT THAT PHOTON PASSED THROUGH TO REACH THE SCREEN

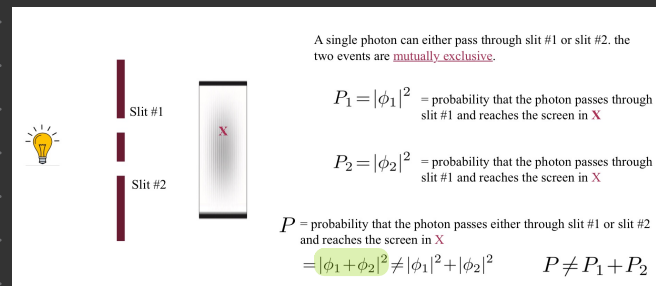


BOTH EVENTS ARE MUTUALLY EXCLUSIVE

THE ESSENTIAL DIFFERENCE BETWEEN 2 COIN EXPERIMENT (SUM OF PROB.) AND TWO-SLIT EXPERIMENT (SUM OF AMPLITUDES) IS **INFORMATION**

THE TWO POSSIBLE PATHS THAT THE PHOTON CAN TAKE CANNOT BE DISTINGUISHED

THE QUANTUM RULE FOR ADDING PROBABILITY AMPLITUDES APPLIES WHEN THE SYSTEM IS **INFORMATIONALLY ISOLATED**

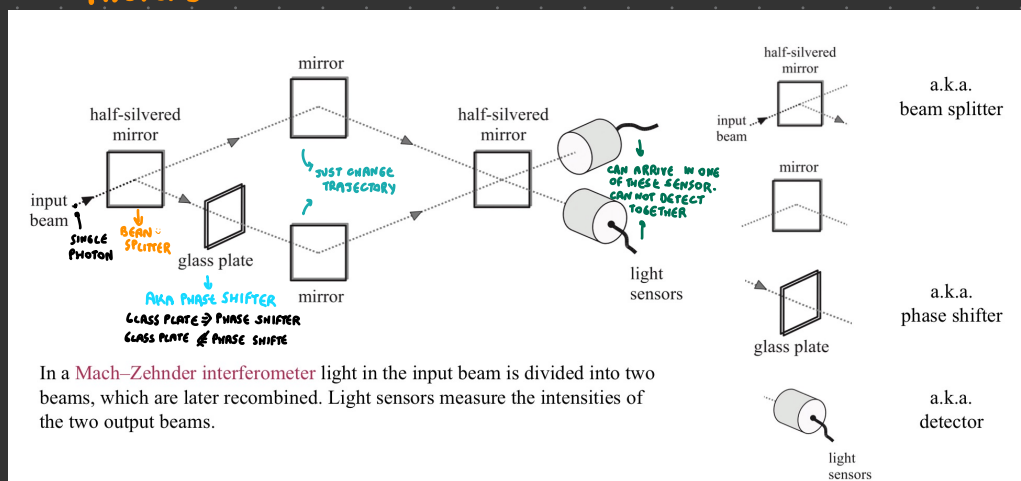


BASIC QUANTUM SYSTEM

- **PHOTONS**: VERY EASY TO MANIPULATE AND MAINTAIN QUANTUM PROPERTIES FOR LONG TIME
- **SPINS 1/2**: IN ELECTRONS
- **TWO-LEVEL ATOMS**

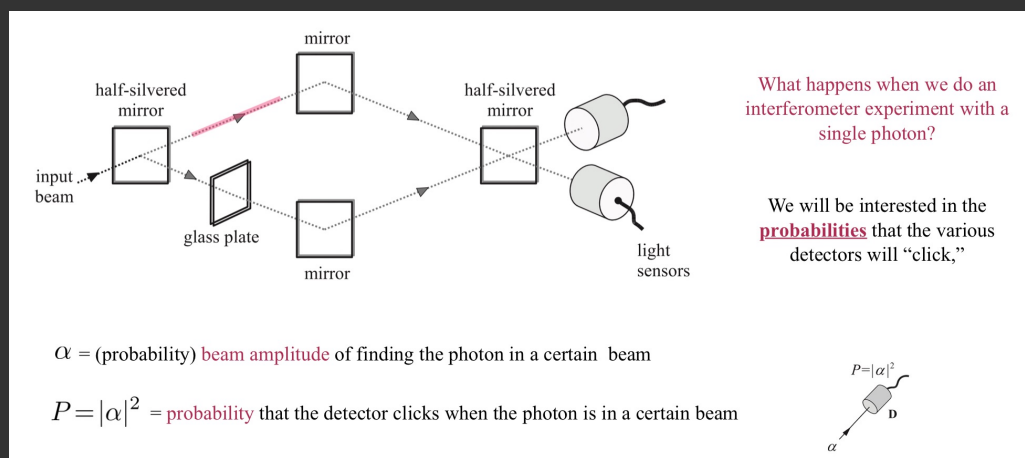
QUANTUM PROPERTIES NECESSITATES A SYSTEM THAT IS INFORMATIONALLY ISOLATED. IF A SYSTEM INTERACT VERY EASILY IS NOT INFORMATIONALLY ISOLATED

PHOTONS



MACH-ZEHNDER INTERFEROMETER LIGHT IN THE INPUT BEAM IS DIVIDED INTO TWO BEAMS, WHICH ARE LATER RECOMBINED. LIGHT SENSORS MEASURE THE INTENSITIES OF THE TWO OUTPUT BEAM

- WHAT HAPPEN WHEN WE DO AN INTERFEROMETER EXPERIMENT WITH A SINGLE PHOTON?
- WE WILL BE INTERESTED IN THE **PROBABILITIES** THAT THE VARIOUS DETECTOR WILL "CLICK"



α = **PROB. BEAM AMPLITUDE** OF FINDING THE PHOTON IN A CERTAIN BEAM

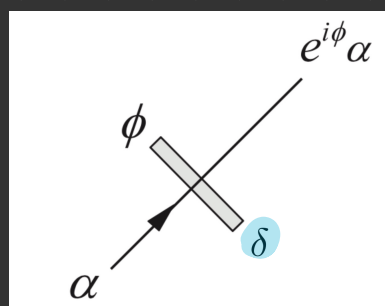
$P = |\alpha|^2$ = **PROBABILITY** THAT THE DETECTOR CLICKS WHEN THE PHOTON IS IN A CERTAIN BEAM

COMPLEX AMPLITUDE

• **BEAM AMPLITUDES**: ARE DESCRIBED AS COMPLEX NUMBER

$$\alpha = |\alpha| e^{i\theta} = |\alpha| (\cos \theta + i \sin \theta)$$

• THE PHASE OF THE PROBABILITY AMPLITUDE CAN BE CHANGED USING A **PHASE SHIFTER**

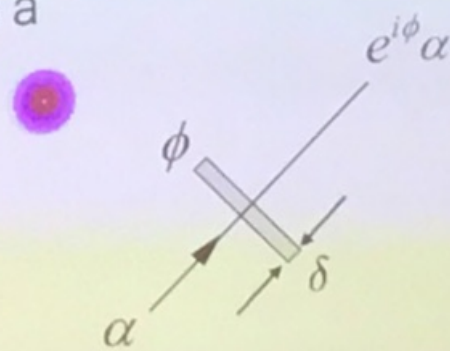


THIS CAN BE IMPLEMENTED BY MEANS OF A TRANSPARENT GLASS SLIDE WITH A PROPER THICKNESS δ

$$e^{i\phi} \alpha = |\alpha| e^{i(\theta + \phi)}$$

Exercise 2.2

- Suppose δ is the smallest thickness of glass that produces a phase shift of π . What is the phase shift if the glass plate has a thickness of $\delta/5$?



- What is the next-to-smallest thickness $\bar{\delta}$ that produces the same change in phase of δ ? What phase shift would be produced by a plate of thickness $\bar{\delta}/5$?

1. THICKNESS δ

$$\lambda \rightarrow \delta \rightarrow \alpha e^{i\pi} \quad \phi = \pi$$

THICKNESS $\frac{\delta}{5}$

$$\lambda \rightarrow \frac{\delta}{5} \rightarrow \alpha e^{i\pi/5}$$

$$\phi = k \delta$$

DEPENDS ON MATERIAL

SUPPOSE I WANT A PHASE SHIFT OF 2π ...

$$\lambda \rightarrow 2\delta \rightarrow \alpha e^{i2\pi}$$

... I SHOULD HAVE A THICKNESS OF 2δ

2.

$$e^{i(\pi+2\pi)} = e^{i\pi}$$

$$\bar{\delta} = \delta + 2\delta$$

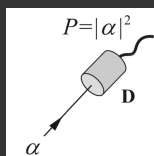
$$\phi = \pi + 2\pi = 3\pi$$

$$\bar{\delta} = 3\delta$$

$$\frac{\bar{\delta}}{5} = \frac{3}{5}\delta$$

$$\frac{\delta}{5} \rightarrow \frac{3}{5}\delta \rightarrow \frac{3}{5}\pi$$

DETECTORS ARE **INSENSITIVE TO THE PHASE** OF THE PROBABILITY AMPLITUDE



$$P = |\alpha|^2 = |e^{i\phi} \alpha|^2$$

- LET'S CONSIDER THE CASE IN WHICH THE PHOTON CAN BE FOUND IN TWO POSSIBLE BEAMS, "UPPER" AND "LOWER"

"UPPER" $\rightarrow$ SITUATION A: PHOTON IN UPPER BEAM

"LOWER" $\rightarrow$ SITUATION B: PHOTON IN LOWER BEAM

- THE TWO SITUATIONS CAN BE REPRESENTED USING TWO VECTORS CONTAINING THE CORRESPONDING BEAM AMPLITUDES

SIT. A $\begin{pmatrix} e^{i0} \\ 0 \end{pmatrix}$ $\leftarrow$ "UPPER" BEAM AMPLITUDE

SIT. A $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

SIT. B $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

PRINCIPLE OF SUPERPOSITION

- MORE IN GENERAL, ONE CAN HAVE OTHER SITUATIONS IN WHICH THE BEAM AMPLITUDES ARE **LINEAR COMBINATIONS** OF THE ASSIGNMENTS FOR A AND B

$$\alpha(\text{SITUATION A}) + \beta(\text{SITUATION B})$$

ARE NOT PROB. OF HAVING PHOTON IN A AND B (ARE COMPLEX NUMBER)

$$\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$P_{UP} = |\alpha|^2$$

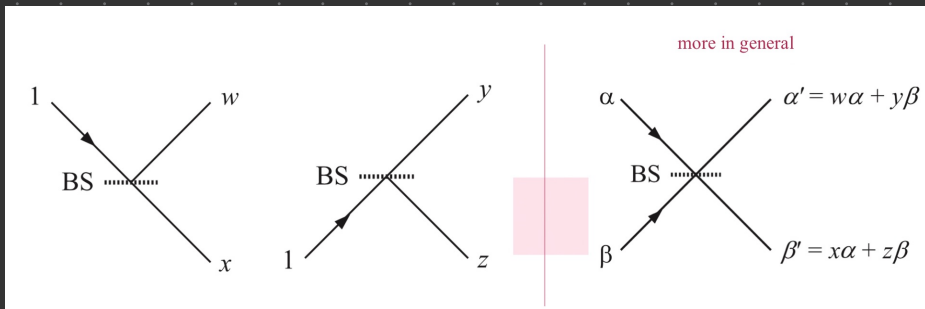
$$P_{LOW} = |\beta|^2$$

NORMALIZATION CONDITION

$$|\alpha|^2 + |\beta|^2 = 1$$

AMPLITUDE VECTOR CORRESPONDING TO A SITUATION IN WHICH THE PHOTON CAN BE SOMETIMES UP AND SOMETIMES DOWN

BEAM SPLITTER TAKES AN INPUT BEAM AND SPLITS IT INTO TWO BEAMS.



$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \begin{pmatrix} w & y \\ x & z \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

THE ACTION OF THE BEAM SPLITTER IS SPECIFIED ONCE w, y, x AND z ARE KNOWN

COMPLEX NUMBER AND NOT-INDEPENDENT FOR SOME CONDITION THAT MUST BE SATISFIED

1. $|\alpha'| + |\beta'|^2 = 1$ (NORMALIZATION)

2. $|w|^2 + |x|^2 = 1$ (ENERGY CONSERVATION)

3. $|y|^2 + |z|^2 = 1$ (ENERGY CONSERVATION)

THERE ARE ALSO ADDITIONAL CONDITIONS THAT DEPEND ON THE WAY IN WHICH THE BEAM SPLITTER IS IMPLEMENTED

50-50 BEAM SPLITTER

THE BEAM AMPLITUDE FROM ONE INPUT PORT IS EVENLY SPLIT IN THE TWO OUTPUT PORTS. THIS CAN BE OBTAINED WITH A HALF-SILVERED MIRROR

$$\begin{pmatrix} w & y \\ x & z \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

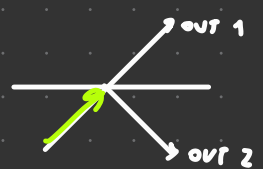
$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$P_{UP} = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

$$P_{LOW} = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

FOR BEAM



$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$P_{UP} = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

$$P_{LOW} = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

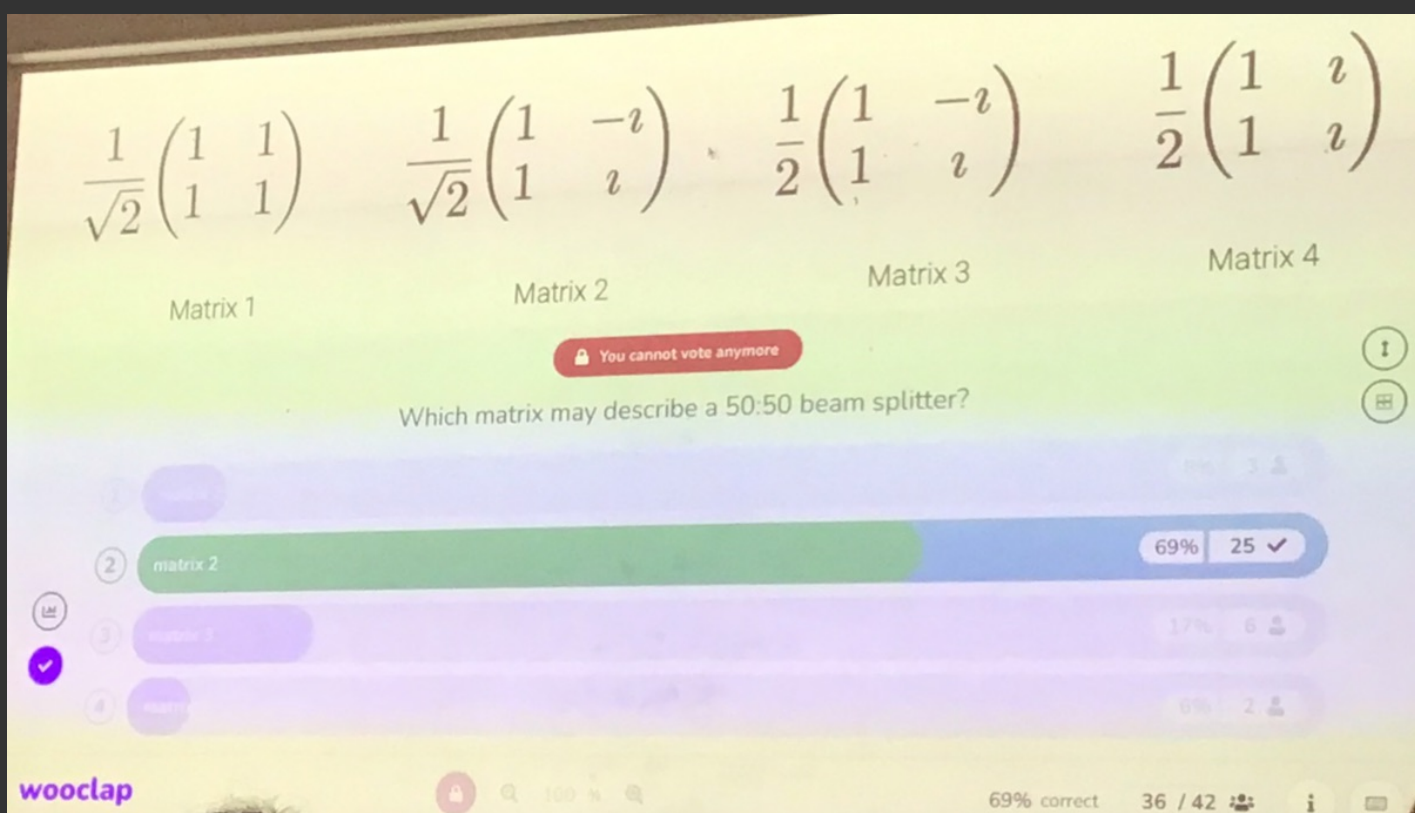
* THE NEGATIVE SIGN IN THE LOWER-RIGHT (2) MATRIX ELEMENT MEANS THAT WHEN THE LOWER INPUT BEAM IS REFLECTED, IT UNDERGOES A PHASE SHIFT OF π . THIS IS INDICATED IN THE SCHEME

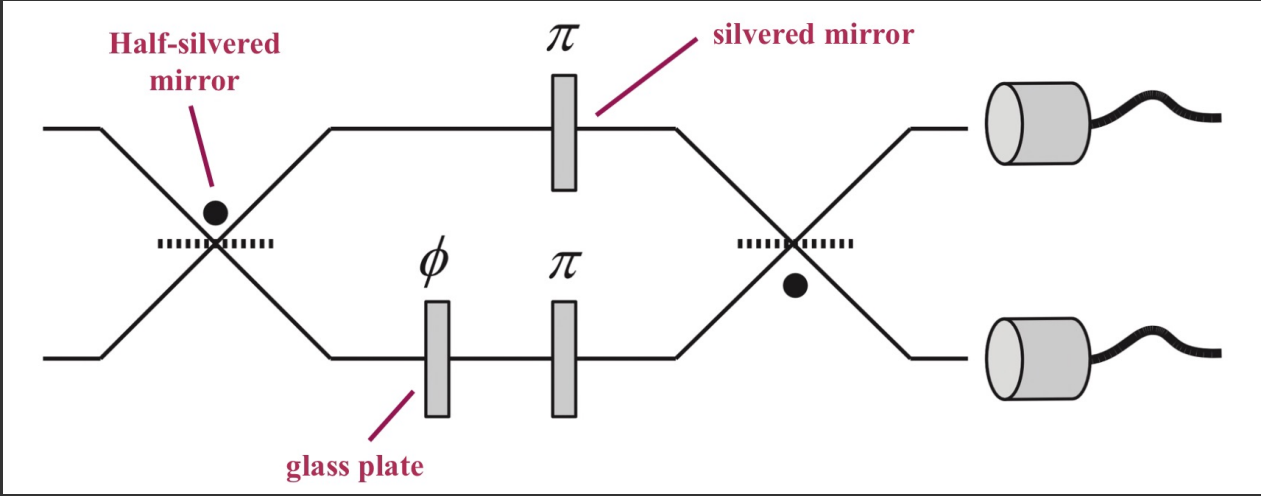
FIND NORM

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha + \beta \\ \alpha - \beta \end{pmatrix} = \left(\frac{\alpha + \beta}{\sqrt{2}} \right) \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} + \left(\frac{\alpha - \beta}{\sqrt{2}} \right) \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

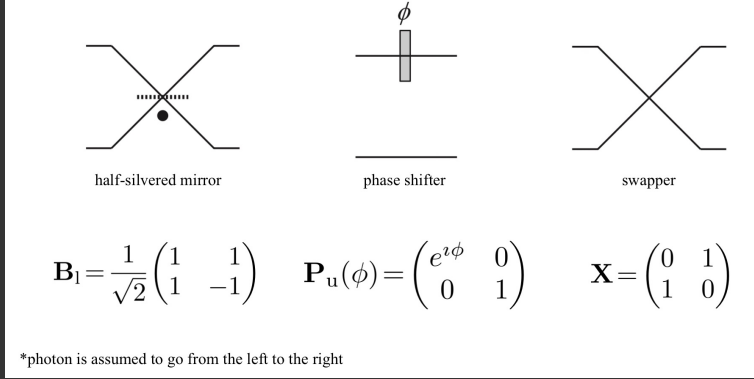
$$= \frac{|\alpha|^2}{2} + \frac{|\beta|^2}{2} + \frac{\alpha\beta^* + \beta\alpha^*}{2} + \frac{|\alpha|^2}{2} + \frac{|\beta|^2}{2} - \frac{\beta\alpha^* + \alpha\beta^*}{2} =$$

$$= |\alpha|^2 + |\beta|^2 = 1$$





AT ANY STAGE OF A MACH-ZEHNDER INTERFEROMETER THE PHOTON **MAY BE** IN ONE OF TWO POSSIBLE BEAMS. WE HAVE DRAWN OUR DIAGRAM, SO THAT ONE BEAM IS "UPPER" BEAM AND ONE IS "LOWER" BEAM



$$\begin{aligned} \textcircled{1} \quad & \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right] = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \textcircled{2} \quad & \begin{pmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{-i\phi} & 0 \\ 0 & 1 \end{pmatrix} = \begin{bmatrix} e^{i\phi} e^{-i\phi} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \textcircled{3} \quad & \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

DET = 1

$A^E \quad A^T \quad A^T$ TRANSPOSE
 $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 1I$$

$(A^T)^* = A^T$
 $|det(A)| = 1$

$AA^\dagger = A^\dagger A = 1I$ **UNITARY MATRIX**

$$B_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad P_u(\phi) = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

These 2x2 matrices correspond to **different** linear optical elements, but they all have something in common. What?



Any physically possible linear optical element in a two-beam interferometer is represented by a 2×2 **unitary matrix**

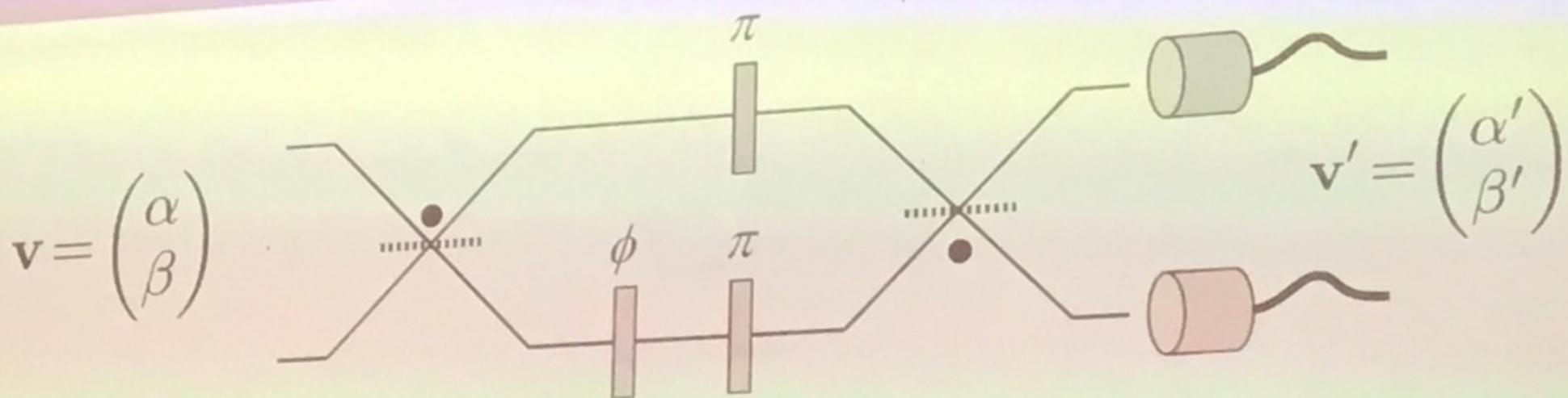
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These 2x2 matrices correspond to **different** linear optical elements, but they all have something in common. What?



Any physically possible linear optical element in a two-beam interferometer is represented by a 2×2 **unitary matrix**

Matrix method



$$\mathbf{v}' = \mathbf{B}_l \mathbf{P}_u(\pi) \mathbf{P}_l(\pi) \mathbf{P}_\phi(\phi) \mathbf{B}_u \mathbf{v}$$

$$\mathbf{v}' = \mathbf{B}_l \mathbf{P}_u(\pi) \mathbf{P}_e(\pi) \mathbf{P}_e(\phi) \mathbf{B}_u \mathbf{v}$$

$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^{i\pi} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{aligned} & \begin{matrix} \text{X} \end{matrix} \downarrow \begin{matrix} \text{X} \end{matrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ e^{i\phi} & e^{i\phi} \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\ &= -\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} +1 & 0 \\ 0 & +1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ e^{i\phi} & e^{i\phi} \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \end{aligned}$$

MULTIPLY BY
THE IDENTITY
MATRIX DOES
NOT CHANGE
ANYTHING

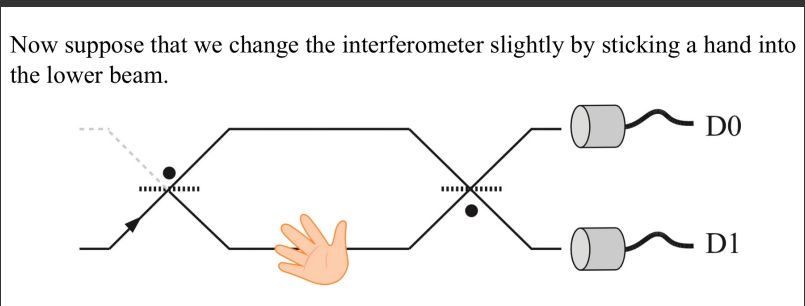
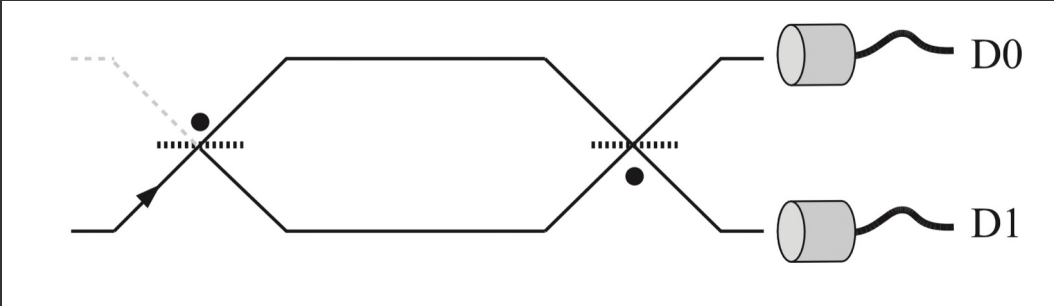
$$= -\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ e^{i\phi} & e^{i\phi} \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$= -\frac{1}{2} \begin{bmatrix} -1 + e^{i\phi} & 1 + e^{i\phi} \\ -1 - e^{i\phi} & 1 - e^{i\phi} \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$\mathbf{T}$

$$\mathbf{T} \cdot \mathbf{T}^\dagger = \mathbf{T} (\mathbf{T}^\dagger)^\dagger = (\mathbf{T}^\dagger)^\dagger \mathbf{T} = \mathbf{1I}$$

- TESTING BOMBS**
- THE COMPONENTS OF AN INTERFEROMETER DO NOT REGISTER THE PASSAGE OF A PHOTON
 - THUS, THE PHOTON REMAINS INFORMATIONALLY ISOLATED. THIS IS WHY THE BEAMS EXHIBIT INTERFERENCE



$$p_{D0} = |\alpha'|^2$$

$$p_{D1} = |\beta'|^2$$

$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{2} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

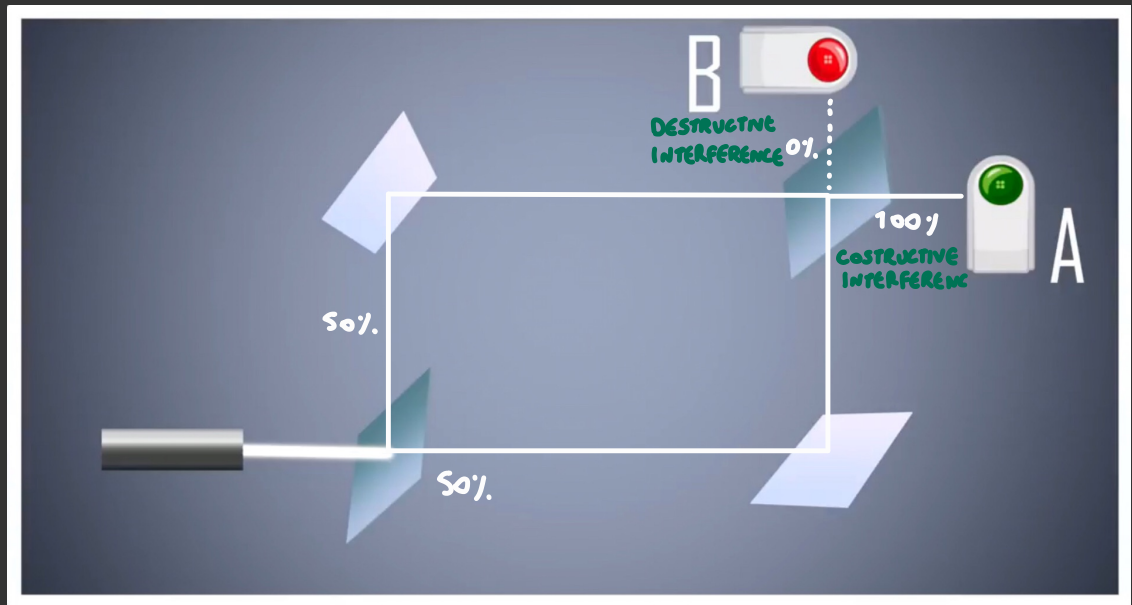
INTERFERENCE

$p_{D0} = |1|^2 = 1$
 $p_{D0} = 1$
 $p_{D1} = 0$

Outcome	Probability
Photon in D0	1
Photon in D1	0

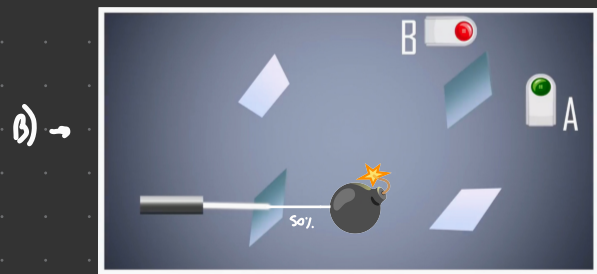
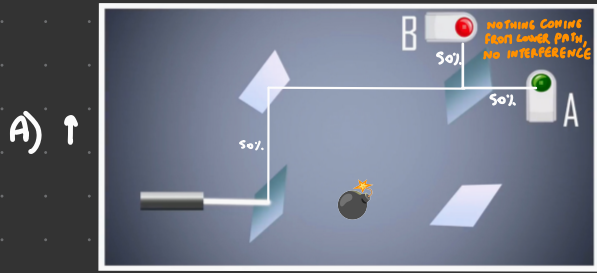
Outcome	Probability
Photon in D0	1/4
Photon in D1	1/4
Photon hits	1/2

- WHETHER OR NOT A WORKING BOMB ACTUALLY DETECTS A PHOTON IN A GIVEN TRIAL, ITS FINAL STATE (INTACT OR EXPLODED) PROVIDES A RECORD OF WHICH BEAM THE PHOTON HAS TRAVERSED
- THIS MEANS THAT THE PHOTON IS NOT INFORMATIONALLY ISOLATED IN THE APPARATUS, AND SO THERE CAN BE NO INTERFERENCE BETWEEN THE BEAMS



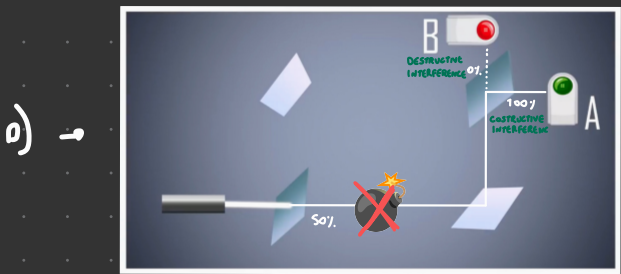
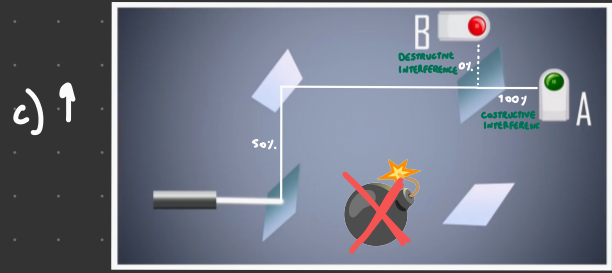
GOOD BOMB (WORKING)

BAD BOMB (NOT WORKING)



50% BLOWS UP
25% IN A
25% IN B

IS THE BOMB WORKING?
• DETECT IN A: UNSURE
• DETECT IN B: BOMB WORKING BUT NOT EXPLODE



0% BLOWS UP
100% IN A
0% IN B